

NETWORK LAYER

Module 3

Block Allocation:

- For CIDR to work properly, **two restrictions** must be **followed** when **allocating an address block**:
 1. The **number of addresses** (N) must be a **power of 2**. **For example**: 2, 4, 8, 16, 32, 64, etc. This is because the formula is: $N = 2^{32-n}$, Therefore, if N is not a power of 2, the value of n will not be an integer.
 2. The **addresses in the block** must be **contiguous (consecutive)** and the **first address** of the block must be **divisible by N**, where N is the total number of addresses in the block. **For example**, if a block contains 16 addresses, its first address must be divisible by 16. This is because the first address is obtained by taking the prefix and adding 0s for the remaining (32-n) bits.

For example: An ISP has requested a block of 1000 addresses. Since 1000 is not a power of 2, 1024 addresses are granted. The prefix length is calculated as $n = 32 - \log_2 1024 = 22$. An available block, 18.14.12.0/22, is granted to the ISP. It can be seen that the first address in decimal is 302,910,464, which is divisible by 1024.

Divisibility can be checked by following two methods:

Rule 1: Convert whole octet in decimal

$$18 \times 256^3 + 14 \times 256^2 + 12 \times 256^1 + 0 \times 256^0 = 302,910,464$$
$$302,910,464 / 1024 = 2,95,811$$

Rule 2:

$$\text{Total number of addresses} = 1024 = 2^{10} = 2^2 \times 2^8 = 4 \times 256$$

Take last two octets and divide first with 4 and second with 256, we get $12/4 = 3$, $0/256 = 0$

Subnetting in Classless Addressing:

- More levels of hierarchy can be created using subnetting. An organization (or an ISP) that is granted a range of addresses may divide the range into several subranges and assign each subrange to a subnetwork (or subnet).
- A subnetwork can be divided into several sub-subnetworks. A sub-subnetwork can be divided into several sub-sub-subnetworks, and so on.

- Assume the total number of addresses granted to the organization is N , the prefix length is n , the assigned number of addresses to each subnetwork is N_{sub} , and the prefix length for each subnetwork is n_{sub} . Then the following steps need to be carefully followed to guarantee the proper operation of the subnetworks.
 - The number of addresses in each subnetwork should be a power of 2.
 - The prefix length for each subnetwork should be found using the following formula:
$$n_{sub} = 32 - \log_2 N_{sub}$$
 - The starting address in each subnetwork should be divisible by the number of addresses in that subnetwork.

Problem: Let one of the address of the block is 192.126.120.98/26. If this block is divided into four equal subnets. Calculate the range of each sub-block (subnet).

Solution: Do it in class.

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Answer:

- Range of block: 192.126.120.64/26 ---- 192.126.120.127/26
- Range of 1st Subnet: 192.126.120.64/28 ---- 192.126.120.79/28
- Range of 2nd Subnet: 192.126.120.80/28 ---- 192.126.120.95/28
- Range of 3rd Subnet: 192.126.120.96/28 ---- 192.126.120.111/28
- Range of 4th Subnet: 192.126.120.112/28 ---- 192.126.120.127/28

References:

Data Communications and Networking by Behrouz A. Forouzan.



THANK YOU